

Charan Vardham

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EDUCATION

University of Wisconsin Madison

Madison, WI

Bachelor of Science in Computer Science, Minor in Mathematics

Aug. 2024 – May 2028

- **Relevant Courses:** Data Structures, Algorithms, Discrete Mathematics, Linear Algebra, Computer Security

EXPERIENCE

AI/ML Intern

May 2025 – Aug 2025

IQVIA

Remote

- Built a scalable RAG-based medical chatbot to minimize hallucinations and improve factuality processing over 200MB of clinical documents
- Implemented document ingestion, text chunking, embedding generation, and retrieval pipelines achieving 90% retrieval accuracy with MongoDB-backed chat history
- Worked on a separate voice-enabled chatbot project, integrating NVIDIA Riva APIs for real-time STT/TTS with an LLM backend

Undergraduate Student Researcher

Aug 2025 – Present

Case Western Reserve University

Cleveland, OH

- Designed and trained deep learning CNN models in PyTorch on 5000+ CT scan slices for cross-kernel image reconstruction, optimizing model accuracy using PSNR and SSIM evaluation metrics
- Implemented GPU-accelerated training pipelines on CUDA hardware, applying knowledge distillation and GAN-based architectures (pix2pix, CycleGAN) for domain transfer in medical computer vision tasks
- Conducted applied research into hardware-software integration for scalable deep learning inference; preparing findings for publication

Open Source Contributor

May 2026 – Present

PyTorch

Remote

- Fixed 3 bugs in PyTorch Inductor's `torch.compile()` path spanning `IndexPropagation`, `sympy.Min/Max`, and `torch.clamp` bounds handling
- Improved `torch.autograd.anomaly_mode` docs and resolved B008 flake8-bugbear violations across the codebase

xLab Intern

Jan 2026 – May 2026

Weatherhead School of Management, CWRU

Cleveland, OH

- Developing an AI-powered case study platform using LLMs to simulate real-world business decision-making
- Building a full-stack application with Next.js (TypeScript), Tailwind CSS, PostgreSQL, and Prisma

PROJECTS

GPT-2 Pretraining | *PyTorch, Python, CUDA, SLURM*

February 2026

- Trained GPT-2 (124M parameters) from scratch on 10B tokens using NVIDIA H100 GPUs, achieving final loss of 3.0
- Optimized training with bfloat16 mixed precision and batch scheduling, achieving 260K tokens/sec throughput

Real-Time Credit Card Fraud Detection | *Python, TypeScript, XGBoost, AWS, Docker*

June 2026

- Built an end-to-end ML fraud detection system using a stacking ensemble model trained on the IEEE-CIS dataset and deployed in a Docker container on ECS Fargate via SST v3
- Implemented a Next.js 15 inference API (predict, transaction history, health check) with TypeScript types shared across the stack, writing predictions to DynamoDB on every request
- Designed CI/CD on GitHub Actions with TypeScript type-checking, Python linting, and automated SST deployment to AWS

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, R, HTML/CSS

Frameworks / Libraries: React, Node.js, Next.js, FastAPI, Material-UI, Pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch, TensorFlow, PyTorch Lightning, Hugging Face Transformers

Developer Tools / Deployment: Git, Docker, VS Code, IntelliJ, PyCharm, Visual Studio, REST APIs, AWS, CI/CD, Jenkins, Kubernetes

Testing / ML Skills: Unit testing, Data preprocessing, Feature engineering, Model evaluation metrics (PSNR, SSIM), RAG pipelines, LLM integration